



Seaglass Globe™

The Agent's Presentation Surface — a Globe That Can Be Flown by Software

A precision WGS-84 virtual globe with a native Model Context Protocol server: one POST `/mcp` endpoint on the running application, loopback-bound and bearer-protected in the suite install. Clients attach to a running globe rather than launching one — the user owns the window, and the agent joins it. The surface covers the camera, layers, projection, screenshots, scene entities, reports and raster overlays, military symbology and scenarios, and the full compute bridge: ForgeGIS™ operations, pipelines and batch runs; ForgeData™ catalog, extraction, elevation and publish. Results come back as things on the globe, not descriptions of them.

/mcp

one endpoint, one transport

MCP Streamable HTTP on the running app, loopback only; a stdio adapter for clients, ForgeMind™ included, that speak stdio

Scene

sent with the question

new in 1.2.5: the Assistant sends what is on the globe — entities with their symbol codes, draped overlays, the scenario clock — as structured context

Bounded

every tool call

a wedged UI thread or child engine returns an error to the caller instead of hanging the agent; a panicking tool is caught, and the window stays up

Attach, don't spawn

The stdio adapter proxies to a running globe's `/mcp`; the desktop app serves that endpoint itself. An agent-launches-the-app model is wrong for an interactive viewer, so the design is the other way round: the port is bound before the window and the GPU come up, and readiness is observable — `tools/list`, then `list_ops` for ForgeGIS, then `list_datasets` for ForgeData. A tool called `early` answers warming, not an error.

Results land on the globe

Every compute result is routed by shape. A summary goes to the report panel; a GeoJSON FeatureCollection becomes a scene entity; a raster handle is draped, reprojected by ForgeGIS, never guessed. A partial draw is reported as “drew N of M”, never as success — and in 1.2.5 a large result from a projected source is declined with a notice rather than drawn wrongly. `analyze_area` is the one-shot `best-dataset` → `extract` → `compute` → `drape` path.

Answers that survive a restart

`publish_scene_entity` records an answer the operator can reference by name in later turns — site an HLZ as “Site 1”, then “route from Site 1”. When the ForgeData catalog is up it is written through, so the entity comes back after a restart with the same id. New in 1.2.5, `select_scene_entity` selects one for anchor editing, and `list_renderables` lists what the agent can toggle — feeds, agent geometry, heatmaps — with what each draws right now. `undo_last` undoes the last entity change from any source.

WHO THIS IS FOR

Teams building on ForgeMind™

The window you type into is the map that answers: ask in the Globe's Assistant and the answer is drawn on the globe; ask in ForgeGIS Studio™ and it lands on Studio's map. It is the same agent either way. The Assistant ships closed — View → Panels → Assistant docks it — and its bearer rides each request, never a client default.

MCP-native agent platforms and orchestrators

Anything that speaks MCP over HTTP can drive the running globe — initialize, ping, tools/list, tools/call. Tool descriptions are hand-written agent-facing prose, because the description is what decides whether the tool gets picked. Every object-shaped result is also sent as structuredContent, so a client never has to pick a text block by position.

Builders of geospatial and exercise workflows

Fly the camera, load a raster into an engine-side handle, run an operation or a saved pipeline against it, drape the result; browse the catalog, pull an extract, publish an answer back. New in 1.2.5, place MIL-STD-2525 symbols and tactical graphics, load and drive scenarios, and stream records into the scenario library over the same port. It is the same globe the operator sees, and a shape they draw by hand is the same kind of entity the agent places.

Fire-and-queue, by design. Compute tools return {"ok": true, "queued": true} immediately; the overlay lands a moment later. An agent that expects a synchronous result will look at an empty globe and conclude failure — poll list_handles or list_scene_entities, or read the report. Handles are opaque ids for rasters in the engine's memory, cleared when a child engine restarts.

Read-back gaps, stated. list_renderables now lists the toggleable renderables, but agents still can't read imagery or vector-file layer state back (list_layers), and pick_at and tour or bookmark tools are not on the surface yet. The loopback bind and the per-install bearer are the boundary; a hand-launched globe with no token configured has none, so never expose the port without setting one. Keep the window open and unminimised while an agent works: a covered or minimised globe stops drawing, and its requests time out.

Build on the Globe

The Technical Brief covers the MCP surface, the compute and catalog bridges, the entity and scenario model, the digital-twin wire, and deployment; the generated tool reference ships in the suite documentation. Seaglass Globe is an opt-in component of the Forge suite, installed by the Seaglass Control Center™, and is not sold separately. The Forge suite is commercially licensed; evaluation builds are available to qualified teams.

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